

Continuous Random Variable

discrete R.V

x	0	-	-	-	n
p(x)					

$$\textcircled{1} \text{ mean} = E(x) = \mu = \sum x p(x)$$

$$\sum p = 1$$

$$\begin{aligned} \textcircled{2} \text{ variance} &= \sigma^2 = \sum x^2 p(x) - \mu^2 \\ &= \sum (x - \mu)^2 p(x) \end{aligned}$$

Binomial & Geometric

Continuous R.V

Continuous distributions

① Exponential dist

② Normal distribution $\equiv$ Z-dist

↳ Standard N.D

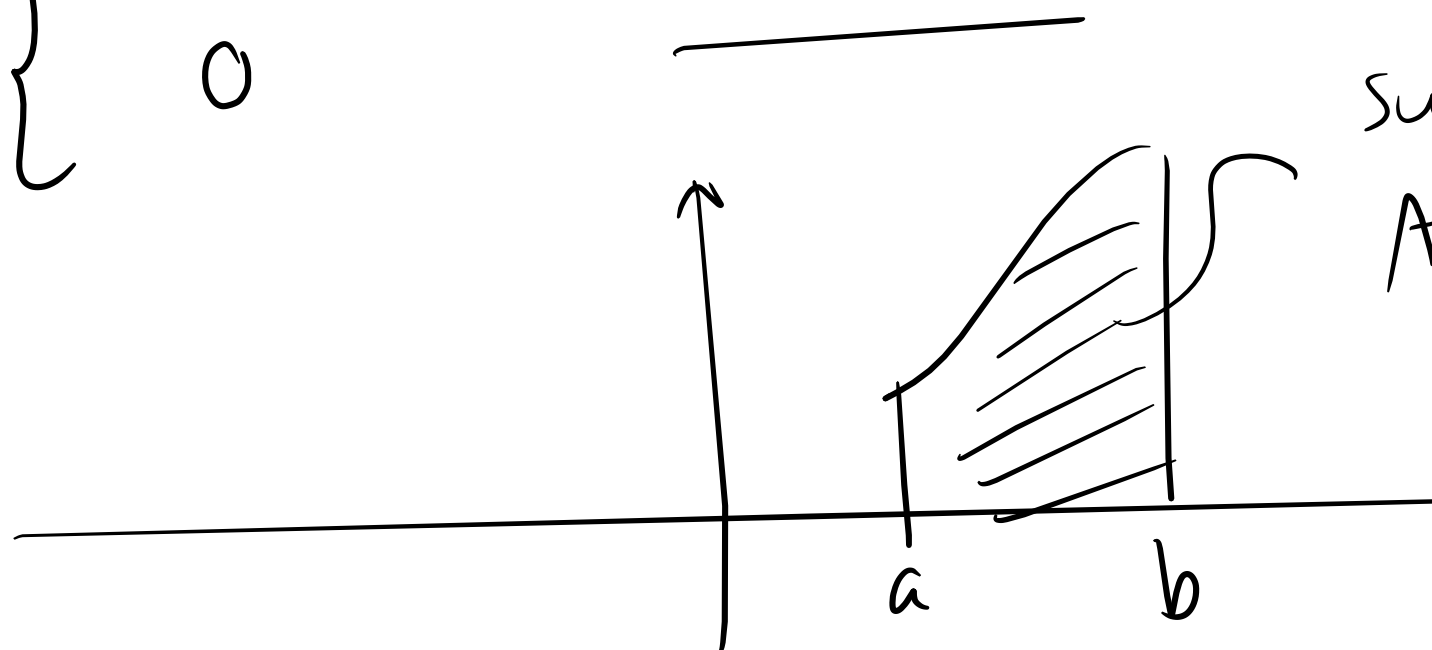
σ, μ

③ t-distribution

$\left\{ \begin{array}{l} Z-D \\ t \rightarrow \\ C.I \\ H_0, H_1 \end{array} \right.$

$$P(x) = \begin{cases} P(x) \\ 0 \end{cases}$$

$$a < x < b$$



Sum of
Area = 1

$$\int_{-\infty}^{\infty} P(x) dx = 1$$

$$\textcircled{1} \int_{-\infty}^{\infty} p(x) dx = 1$$

$$\textcircled{2} \text{mean} = \mu = E(x) = \int_{-\infty}^{\infty} x p(x) dx$$

$$\begin{aligned} \textcircled{3} \sigma^2 &= \int_{-\infty}^{\infty} x^2 p(x) dx - \mu^2 \\ &= \int_{-\infty}^{\infty} (x - \mu)^2 p(x) dx \end{aligned}$$

$$\textcircled{4} \text{CPF}$$

$$F(x) = \int_{-\infty}^x p(x) dx$$

○ Is the function

$$f(x) = \begin{cases} 0, & x < -1, \\ x^2, & -1 \leq x \leq 1, \\ 0, & x > 1, \end{cases}$$

a probability density function?

$$\int_{-1}^1 f(x) dx = \int_{-1}^1 x^2 dx = \left. \frac{x^3}{3} \right|_{-1}^1 = \frac{1}{3} [1^3 - (-1)^3] \\ = \frac{1}{3} [2] = \frac{2}{3} \neq 1$$

$f(x)$ is Not p.m.f
لأنه لا يساوي 1

• Is the function

$$f(x) = \begin{cases} 0, & x < -1, \\ 1.5x^2, & -1 \leq x \leq 1, \\ 0, & x > 1, \end{cases}$$

a probability density function?

$$\begin{aligned} \int_{-1}^1 1.5x^2 dx &= 1.5 \int_{-1}^1 x^2 dx \\ &= 1.5 \left. \frac{x^3}{3} \right|_{-1}^1 = \frac{1.5}{3} [1^3 - (-1)^3] \\ &= 1.5 [2] \\ &= \frac{3}{2} (2) \left(\frac{1}{3} \right) \\ &= 1 \end{aligned}$$

Yes

p.m.f

Find the value of c so that

$$f(x) = \begin{cases} 0, & x < 0, \\ cxe^{-x^2}, & x \geq 0, \end{cases}$$

is a PDF.

$$\int_0^{\infty} c x e^{-x^2} dx = 1$$

$$c \int_0^{\infty} x e^{-x^2} dx = 1$$

$$\begin{aligned} \int x e^{-x^2} dx &= -\frac{1}{2} \int -2x e^{-x^2} dx \\ &= -\frac{1}{2} e^{-x^2} \Big|_0^{\infty} \\ &= -\frac{1}{2} [e^{-\infty} - e^0] \\ &= -\frac{1}{2} [0 - 1] = \frac{1}{2} \end{aligned}$$

$$\int e^{g(x)} g'(x) dx = e^{g(x)}$$

$$\frac{1}{3} \int 3 e^{3x} dx = \frac{1}{3} e^{3x}$$

$$\frac{1}{2} = \frac{1}{2} \therefore \frac{c}{2} = 1 \quad \boxed{c = 2}$$

$$f(x) = \begin{cases} 0 & x < 0 \\ 2x e^{-x^2} & 0 \leq x \end{cases}$$

mean value $= \mu = \int_0^{\infty} x f(x) dx = \int_0^{\infty} x (2x e^{-x^2}) dx$

$$= 2 \int_0^{\infty} x^2 e^{-x^2} dx =$$

$$y = x^3 + C$$

$$\frac{dy}{dx} = 3x^2$$

$$\int dy = \int 3x^2 dx$$

$$y = \frac{3x^3}{3} + C$$

◦ Similarly, one can see that the function

$$f(x) = \begin{cases} 0 & , \quad x \leq 0, \\ 3x^2 e^{-x^3} & , \quad x > 0, \end{cases}$$

$$\begin{aligned} \int_0^{\infty} 3x^2 e^{-x^3} dx &= -e^{-x^3} \Big|_0^{\infty} \\ &= -[e^{-\infty} - 1] \\ &= -[0 - 1] = 1 \end{aligned}$$

p.m.f

mean = ??
G² = ??

$$P(X \geq 3) = \int_3^{\infty} 3x^2 e^{-x^3} dx$$

$$\int x^2 e^{-x} dx$$

↓

diff ↓

x^2	⊕	e^{-x}
$2x$	⊖	$-e^{-x}$
2	⊕	e^{-x}
0		$-e^{-x}$

↓ Integ

$$\boxed{\int -x^2 e^{-x} - 2x e^{-x} - 2e^{-x}}$$